

Indian Healthcare Patient Records Analysis — SQL Analysis

Project Overview

This SQL analysis explores the Indian Healthcare Patient Records dataset to identify patient demographics, disease patterns, treatment trends, hospital performance, insurance coverage, and health indicator variations. The analysis was performed on the cleaned table: `Indian_Healthcare_Cleaned`

1. Dataset Exploration

```
SELECT *  
FROM Indian_Healthcare_Cleaned;
```

2. Total Number of Patients

```
-- 1. total patient count  
SELECT COUNT(*) AS total_patient  
FROM Indian_Healthcare_Cleaned ;
```

```
i total_patient
```

```
100000
```

3. Gender-wise Patient Distribution

```
8 SELECT gender , COUNT(*) AS total_patient  
9 FROM Indian_Healthcare_Cleaned  
10 GROUP BY gender  
11 ORDER BY total_patient DESC ;
```

```
i gender
```

```
total_patient
```

```
Male
```

```
51013
```

```
Female
```

```
48494
```

```
Other
```

```
493
```

4. Disease-wise Patient Distribution

```
SELECT primary_diagnosi , COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY primary_diagnosi
ORDER BY total_patient DESC ;
```

primary_diagnosi	total_patient
Type 2 Diabetes	28214
Hypertension	22086
Cardiovascular Disease	11889
Dengue	7910
Chronic Kidney Disease	6972
Asthma/COPD	5981
Malaria	5898
Tuberculosis	5075
Thyroid Disorder	3959
Other	2016

5. Treatment Type Distribution

```
SELECT treatment_type ,
COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY primary_diagnosi
ORDER BY total_patient DESC ;
```

treatment_type	total_patient
Medication	28214
Lifestyle Changes	22086
Antibiotics	11889
Lifestyle Changes	7910
Combination Therapy	6972
Physiotherapy	5981
Medication	5898
Lifestyle Changes	5075
Combination Therapy	3959
Physiotherapy	2016

This helps understand which treatment methods are most frequently used.

6. Region-wise Patient Distribution

```
SELECT region ,
COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY primary_diagnosi
ORDER BY total_patient DESC ;
```

region	total_patient
Delhi	28214
Maharashtra	22086
Karnataka	11889
Andhra Pradesh	7910
Telangana	6972
Tamil Nadu	5981
Puducherry	5898
Tamil Nadu	5075
Puducherry	3959
Karnataka	2016

7. Region-wise Treatment Analysis

```
SELECT region , treatment_type ,COUNT(treatment_type) AS treatment_type_count
FROM Indian_Healthcare_Cleaned
GROUP BY region
ORDER BY treatment_type_count DESC ;
```

region	treatment_type	treatment_type_count
Tamil Nadu	Combination Therapy	18002
Kerala	Surgery	14794
Karnataka	Antibiotics	12049
Andhra Pradesh	Lifestyle Changes	10140
Telangana	Surgery	8975
Uttar Pradesh	Insulin Therapy	8056
Puducherry	Medication	8008
Maharashtra	Lifestyle Changes	8006
Delhi	Medication	6982
West Bengal	Antibiotics	4988

This helps compare treatment preferences and healthcare service patterns across regions.

8. Hospital Type-wise Patient Distribution

```
SELECT hospital_type , COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY hospital_type
ORDER BY total_patient DESC;
```

hospital_type	total_patient
Government	44647
Private	40141
Corporate	15212

This helps understand which types of hospitals handle the highest number of patients.



9. Insurance Coverage Analysis

This query analyzes the number of patients based on their insurance coverage status.

```
SELECT insurance_covere , COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY insurance_covere
ORDER BY total_patient DESC;
```

insurance_covere	total_patient
False	64970
True	35030

This analysis provides insights into healthcare insurance coverage among patients.

10. Disease-wise Health Indicator Analysis

```
SELECT
    primary_diagnosi,
    ROUND(AVG(blood_glucose_mg),2) AS Avg_Blood_Glucose,
    ROUND(AVG(hba1c),2) AS Avg_HbA1c,
    ROUND(AVG(total_cholesterol),2) AS Avg_Cholesterol,
    ROUND(AVG(BMI),2) AS Avg_BMI
FROM Indian_Healthcare_Cleaned
GROUP BY primary_diagnosi
ORDER BY Avg_Blood_Glucose DESC;
```

primary_diagnosi	Avg_Blood_Glucose	Avg_HbA1c	Avg_Cholesterol	Avg_BMI
Hypertension	110.18	5.8	245.05	24.54
Cardiovascular Disease	109.61	5.78	245.24	24.47
Other	106.18	5.56	195.59	24.62
Malaria	105.39	5.58	194	24.59
Dengue	105.19	5.6	195.07	24.48
Asthma/COPD	104.92	5.59	194.45	24.55
Tuberculosis	104.65	5.59	194.59	24.57
Chronic Kidney Disease	104.64	5.61	193.79	24.64
Thyroid Disorder	104.52	5.61	195.56	24.61

This analysis helps identify how average health indicators vary across different diseases.

11. Hospital Type and Treatment Outcome Analysis

```
SELECT hospital_type, treatment_outcom ,
COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY treatment_type
ORDER BY total_patient DESC ;
```

hospital_type	treatment_outcom	total_patient
Government	Improved	14532
Government	Improved	14379
Government	Improved	14342
Government	Recovered	14253
Private	Referred	14225
Government	Referred	14159
Government	Stable	14110

This helps compare treatment outcomes and understand performance patterns across different hospital types.

12. Year-wise Patient Visit Trend

```
SELECT strftime('%Y',visit_date ) AS Month ,
COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY Month
ORDER BY Month ;
```

Month	total_patient
2024	29966
2025	49822
2026	20212

This helps identify changes in patient volume over time.

Note: `strftime('%Y', visit_date)` extracts the year from the visit date. It does not extract the month.

13. Top 5 Occupations by Patient Count

This query identifies the five occupations with the highest number of patients.

```

SELECT occupation ,
COUNT(*) AS total_patient
FROM Indian_Healthcare_Cleaned
GROUP BY occupation
ORDER BY total_patient DESC
LIMIT 5 ;

```

occupation	total_patient
Chiropodist	198
Runner, broadcasting/film/video	193
Sales promotion account executive	190
Recycling officer	189
Chief Financial Officer	189

This helps understand which occupational groups are most represented among the patients.

14. Treatment Outcome Percentage

```

SELECT
    treatment_outcom,
    COUNT(*) AS total_patient,
    ROUND(
        COUNT(*) * 100.0 /
        (SELECT COUNT(*) FROM Indian_Healthcare_Cleaned), 2
    ) AS percentage
FROM Indian_Healthcare_Cleaned
GROUP BY treatment_outcom
ORDER BY percentage DESC;

```

treatment_outcom	total_patient	percentage
Stable	20213	20.21
Referred	20082	20.08
Worsened	20060	20.06
Recovered	19879	19.88
Improved	19766	19.77

15. Disease-wise Average Age

```
1  SELECT
2      primary_diagnosi,
3      ROUND(AVG(age), 2) AS avg_age,
4      MIN(age) AS min_age,
5      MAX(age) AS max_age
6  FROM Indian_Healthcare_Cleaned
7  GROUP BY primary_diagnosi
8  ORDER BY avg_age DESC;
```

primary_diagnosi	avg_age	min_age	max_age
Hypertension	44.46	1	95
Cardiovascular Disease	44.37	1	95
Thyroid Disorder	44.32	1	95
Dengue	44.32	1	95
Chronic Kidney Disease	44.3	1	95
Tuberculosis	44.27	1	95
Asthma/COPD	44.26	1	95
Other	44.21	1	95

16. Hospital Type vs Average Health Indicators

```
1  SELECT
2      hospital_type,
3      ROUND(AVG(blood_glucose_mg), 2) AS avg_blood_glucose,
4      ROUND(AVG(hba1c), 2) AS avg_hba1c,
5      ROUND(AVG(total_cholesterol), 2) AS avg_cholesterol,
6      ROUND(AVG(BMI), 2) AS avg_bmi
7  FROM Indian_Healthcare_Cleaned
8  GROUP BY hospital_type
9  ORDER BY avg_blood_glucose DESC;
```

hospital_type	avg_blood_glucose	avg_hba1c	avg_cholesterol	avg_bmi
Corporate	128.25	6.41	216.41	24.54
Government	127.74	6.41	216.26	24.57
Private	127.46	6.38	215.91	24.55